



In-situ phase separated ionogel electrolytes with enhanced properties for lithium metal batteries

Boluwatife Igbaroola^a, Nicolas Demarthe^a, Patrick Howlett^b, Maria Forsyth^b,
Galyna Kvasha^c, Andriy Kvasha^c, Susan Sananes-Israel^c, Iratxe De Meatza^c, Nicolas Dupré^{a,*},
Jean Le Bideau^{a,*}

^a Nantes Université, CNRS, Institut des Matériaux de Nantes Jean Rouxel, IMN, F-44000 Nantes, France

^b Institute for Frontier Materials (IFM), Deakin University, 221 Burwood Highway, Burwood, Victoria 3125, Australia

^c CIDETEC, Basque Research and Technology Alliance (BRTA), P.º Miramón, 196, 20014 Donostia-San Sebastian, Spain

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ABSTRACT

Ionogel electrolytes offer a promising alternative to conventional liquid electrolytes, combining the thermal stability of solid-state electrolytes with the high ionic conductivity of liquid electrolytes. Phosphonium-based ionic liquids (ILs) exhibit superior electrochemical properties especially at high lithium salt concentrations. Although acrylic acid (AA)-based ionogels demonstrate favorable lithium-ion diffusion, their mechanical limitations hinder practical application. To address this, a novel ionogel electrolyte is developed through copolymerization of AA with 1,6-hexanediol diacrylate (HDDA). This in-situ phase separation process yields a tough, flexible, and self-standing ionogel capable of confining a high amount (85 wt%) of ionic liquid. The resulting material exhibits excellent ionic conductivity ($\sim 2.5 \text{ mS cm}^{-1}$ at 30°C) and thermo-mechanical properties while maintaining the safety standards of solid electrolytes. Furthermore, the ionogel demonstrated promising cycling stability in a LiFePO₄|ionogel|Li metal battery configuration, retaining 97 % of its initial capacity after 60 cycles at 50°C and 0.2C C-rate. The scalability of this ionogel electrolyte is demonstrated through the successful fabrication of lab-scale single-layer pouch cells. These findings position ionogel electrolytes as a promising technology for next-generation lithium batteries.

1. Introduction

While lithium-ion batteries (LIBs) have revolutionized portable electronics due to their high energy density, safety concerns remain a critical issue, particularly in applications like electric vehicles (EVs). Traditional LIB electrolytes are liquid-based and often contain flammable organic solvents. These pose a significant risk in case of battery failure or overheating (Gebert et al., 2023; Kil et al., 2013). Ionic liquids (ILs) offer a promising solution for safer LIBs. Unlike conventional electrolytes, ILs have negligible volatility and do not readily ignite. Additionally, they boast high ionic conductivity, chemical stability, and thermal stability, making them ideal candidates for next-generation LIB electrolytes (MacFarlane et al., 2014; Armand and Tarascon, 2008; Armand et al., 2009).

Despite their advantages in very good ionic conductivity and ability to dissolve most lithium salts, liquid electrolytes are often associated

with dendrite formation during the cycling of these batteries. During charging and discharging cycles, needle-like lithium dendrites can grow on the electrode. These uncontrolled dendrites can pierce through the porous separator to the positive electrode, leading to short circuits and safety issues. This phenomenon is one of the major reasons why lithium metal anodes, which offer high theoretical capacities, haven't been widely implemented in commercial secondary batteries. Solid electrolytes have emerged as a promising solution to address dendrite formation (Mindemark et al., 2018; Commarieu et al., 2018). Polymer-based and ceramic-based solid electrolytes have recently been explored for solid state battery applications (Aidoud et al., 2018; Murugan et al., 2007; Mecerreyes et al., 2024). While these solid materials offer excellent resistance to dendrite growth, they come with limitations. Ceramic electrolytes often suffer from poor electrode/electrolyte interface and high fabrication costs (Murugan et al., 2007; Kang et al., 2015). Polymer electrolytes, on the other hand, may require elevated operating

* Corresponding authors.

E-mail addresses: nicolas.dupre@cnrs-imn.fr (N. Dupré), jean.lebideau@cnrs-imn.fr (J. Le Bideau).

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temperatures (typically within the 50 °C–60 °C range) to function effectively and still have lithium transport numbers well below 1. These investigations pave the way for alternative approaches in the development of solid electrolytes for lithium battery.

One alternative approach to solid electrolytes explores a class of materials called ionogels which offer a unique combination of properties (Brachet et al., 2014; Guyomard-Lack et al., 2014). Ionogels are prepared by confining an IL within a solid host network. These biphasic materials exhibit the best properties of both liquid and solid electrolytes. They present high ionic conductivity (often associated with liquid electrolytes) and thermal stability of ILs, while also showing the solid properties needed to prevent leakage and potentially mitigate dendrite growth (Aidoud et al., 2016). By confining ILs, their dynamics are maintained or can even be enhanced given the reduction of nano-structured aggregates, reduction of glass transition temperatures (T_g) and the lowering of freezing or melting points (Demarthe et al., 2024). While few microscopic studies have been conducted to quantify these dynamics in ionogel electrolytes for LIBs (Marie et al., 2020), the concept shares similarities with highly concentrated ILs, aiming to promote a “hopping-type” lithium-ion transport mechanism at the interface between the IL and the confining network (Girard et al., 2015; Igbaroola et al., 2024).

Ionogels appeared to be promising alternatives for next-generation solid electrolytes in lithium-ion batteries due to their dual functionality, acting not only as electrolytes but also as separators that can suppress undesirable reactions at the electrode/electrolyte interface (Le Bideau et al., 2007; Echelmeyer et al., 2009; Le Bideau et al., 2011). Some previous research had primarily focused on silica-based matrices or cellulose-derived networks, which benefit from high porosity, electrochemical stability, and a large surface area (often enhanced with silica nanofibers), these materials exhibit significant drawbacks (Guyomard-Lack et al., 2014; Buchtová et al., 2014; Tokuda et al., 2004). Specifically, they are prone to mechanical brittleness, poor interfacial contact with electrodes, shrinkage, and challenges associated with scaling up and processing via the sol–gel method. Other studies have investigated PVDF polymer matrices for ionogel synthesis, a process that necessitates the use of toxic solvents such as dimethylformamide (DMF) and N-methyl-2-pyrrolidone (NMP) to dissolve the polymer and improve its physical integration with the ionic liquid (Brachet et al., 2014; Demarthe et al., 2024). This reliance on hazardous solvents underscores the imperative to investigate alternative polymer matrices, such as those based on acrylate, that can eliminate the need for such substances. Another route was proposed with acrylates, as presented below (Aidoud et al., 2018).

In our previous work, we investigated lithium-ion diffusion in both confined (ionogel) and non-confined IL electrolytes (Igbaroola et al., 2024). Our results, focusing on the promising P_{1114} FSI-LiFSI system (Igbaroola et al., 2024), demonstrated that Li^+ diffusion was significantly more efficient within acrylic acid-based ionogels compared to their non-confined counterparts. This was evidenced by a lower activation energy required for diffusion in the ionogels. Also, we observed a higher diffusion coefficient for Li^+ (^7Li) in the ionogel compared to the non-confined IL, particularly at a high salt concentration of 3 M. However, a crucial factor for using polymer electrolytes, especially in battery applications, is their mechanical behavior. Ideally, these materials should be tough and flexible for ease of handling during synthesis, cell assembly, and ultimately, large-scale production. Unfortunately, while acrylic acid-based ionogels exhibited promising lithium-ion diffusion properties, they were also brittle, making them difficult to handle. Furthermore, for battery applications, it is important to minimize the amount of inactive material (dead mass) within the electrolyte. This is because the polymer matrix, although crucial for providing solid-like properties, contributes to the electrochemical performance only through specific transport mechanism occurring at the solid–liquid interface and not only through the bulk of the solid phase. Additionally, there is a limit to the amount of IL that specific polymers can effectively

confine. For instance, acrylic acid-based ionogels can only accommodate up to 80 wt% IL, and exceeding this limit can hinder the polymerization process as the resulting ionogel becomes glue-like and impossible to handle. Acrylates matrices, in contrast, are known for their good polymerization capabilities. For instance, Aidoud et al. synthesized ionogels by confining an impressive 90 wt% of IL in a matrix of copolymer acrylates, demonstrating the possibility to confine even higher amount of IL (Aidoud et al., 2018).

The interplay between the confined IL and the confining polymer matrix plays a critical role in determining the flexibility of an ionogel (Wang et al., 2022). This relationship can be understood through the concept of monomer solubility within the IL. Highly solvated networks, where the monomer is soluble in the IL, exhibit minimal hydrogen bonding between polymer chains. This is because the IL essentially separates the polymer chains, resulting in transparent, soft, and stretchable ionogels (Zhang et al., 2015). Conversely, poorly solvated networks, where the monomer has low solubility in the IL, lead to opaque, stiff and brittle ionogels (Ueki and Watanabe, 2012). Acrylic acid is poorly soluble in the P_{1114} FSI IL, which explains the observed brittleness of acrylic acid-based ionogels – a significant drawback for battery applications that often require flexibility. A recent study by Wang et al. proposed a promising solution: in-situ phase separation for synthesizing tough and stretchable ionogels (Wang et al., 2022). Their method involves copolymerizing a soluble and a non-soluble monomer within the IL. This approach leads to the formation of a unique network containing both poorly solvated (promoting hydrogen bonding) and highly solvated (elastic) domains. The synergistic effect of these domains enhances the overall toughness of the resultant ionogel. Although previous studies have primarily explored this technique for mechanical applications of ionogels, our work aims to evaluate its potential for developing robust ionogels specifically tailored for lithium metal battery applications.

While an acrylic acid matrix which is insoluble in the P_{1114} FSI IL has demonstrated favorable lithium diffusion and electrochemical properties, copolymerization with a polymer matrix that is soluble in the ionic liquid is critical to improving the mechanical properties of the resulting ionogel. Acrylate matrices are promising candidates in this regard, as they not only exhibit excellent electrochemical performance but also possess superior polymerization characteristics. For example, Aidoud et al. (2018) confined 1 M LiTFSI in a $\text{PYR}_{13}\text{TFSI}$ IL electrolyte within a polymer matrix composed of triacrylate and diacrylate formulations, achieving high IL confinement and producing relatively flexible ionogels suitable for micro-battery applications.

The salt concentration within an ionogel plays a crucial role. It directly affects the viscosity of the precursor solution, which in turn influences greatly the mechanical properties of the final ionogel. A study by De Anastro et al. investigated the impact of sodium salt concentration on the performance of UV-cross-linked ionogels designed for all-solid-state sodium batteries (FdZ De Anastro et al., 2019). Their findings revealed that a higher salt concentration within the ionogels improved the sodium transference number and cycling performance. It is worth knowing that while they showed a slightly lower ionic conductivity, the highly concentrated ionogels demonstrated the ability to withstand higher current densities compared to their lower-concentration counterparts. Particularly, studies have shown that phosphonium-based ILs exhibit better electrochemical performances at high lithium salt concentrations (Araño et al., 2020; Girard et al., 2015). This interplay between salt concentration and various performance parameters necessitates careful optimization for the ionogel synthesis.

In this work, we have developed and characterized flexible phosphonium-based copolymer ionogel electrolytes designed for lithium-metal batteries. These solid-state ionogel electrolytes combine the excellent electrochemical performance and stability achieved by the P_{1114} FSI-LiFSI system, recently investigated by our group, with the mechanical toughness and flexibility obtained by the copolymerization approach. These ionogel electrolytes are composed of a cross-linked

acrylic acid (AA) and 1,6-Hexanediol diacrylate (HDDA) copolymer matrix that confines a P₁₁₁₁₄FSI IL with a high concentration (3 M) of LiFSI salt (Fig. 1). The selected monomers were chosen based on their low molecular weight, high density of polymerization functionalities, and the distinct chemical composition of their spacers, particularly the presence of ethoxy (EO) groups. Ethoxy groups are well-known for their role in facilitating lithium-ion conduction in polymer electrolytes based on polyethylene oxide (PEO) (Aidoud et al., 2018). The copolymer ionogel investigated herein is an innovative and promising family of novel solid-state electrolytes cycled in lithium metal configuration with a LiFePO₄ positive electrode.

2. Experimental

2.1. Materials

The monomers (acrylic acid and 1,6-hexanediol diacrylate), photo-initiator (2-hydroxy-2-methylpropiophenone) and the crosslinker (N,N'-methylenebisacrylamide) were purchased from Sigma-Aldrich. The IL, trimethyl(isobutyl)phosphonium bis(fluorosulfonyl)imide: P₁₁₁₁₄FSI (Cytec Industries, > 99.5 %) was purchased from Boron Molecular (Australia) and bis(fluoromethanesulfonyl) imide lithium salt (LiFSI) was obtained from Solvionic.

2.2. Electrolyte solution and ionogel preparation

An electrolyte solution containing 3 M LiFSI in P₁₁₁₁₄FSI was prepared by dissolving 0.5631 g of LiFSI salt in 1 mL of P₁₁₁₁₄FSI at room temperature in an argon filled glove box. A precursor solution was subsequently prepared by mixing the IL electrolyte solution with the AA, HDDA, crosslinker and photo-initiator in the weight ratio (85:6.5:6.5:1:1), respectively. To ensure homogeneous dispersion of monomers within the IL, the mixture was subjected to magnetic stirring for two hours under ambient atmospheric conditions. The precursor solution was then transferred into Teflon molds with dimensions of 12 mm diameter × 300 μm thickness for coin cells or 60 mm × 50 mm × 500 μm thickness for pouch cells. UV irradiation using a UCUBE-365-025 lamp (250 mW cm⁻²) was employed to initiate the

polymerization process, resulting in the formation of a fully confined network within five minutes of irradiation. Finally, the obtained ionogel samples were dried under vacuum at 60 °C for 24 h to remove residual water before any characterization.

2.3. Chemical characterizations

Chemical characterization was studied using Fourier transform infrared spectroscopy (FTIR, Bruker Alpha 1 spectrometer). FTIR measurements in attenuated total reflection (ATR) mode were taken by keeping the measuring probe on the bare sample at room temperature in the glovebox across the wavelength range of 400 – 4000 cm⁻¹.

2.4. Thermal and mechanical characterizations

Thermal properties of the investigated solid electrolyte membranes were studied by differential scanning calorimetry (DSC), thermogravimetric analysis (TGA) and dynamic mechanical analysis (DMA).

TGA measurements were performed in a STA 449 F5 Jupiter (NETZSCH) instrument using a platinum pan with a small amount of sample (~10 mg). The experiments were recorded in dynamic conditions under air, heating at a rate of 10 °C min⁻¹, from 30 °C to 800 °C.

DSC measurements were performed in a DSC Q20 (TA Instruments) Differential Scanning Calorimeter, equipped with two closed aluminum pans (one as reference and the other with a small amount of product, ~10 mg). Thermograms were recorded at 10 °C·min⁻¹, from -150 °C to 120 °C, and repeated for two heating-cooling cycles. The experiments were recorded in dynamic conditions under nitrogen atmosphere and reported data are taken from the second heating cycle.

Mechanical properties of the ionogels were analyzed in compression mode in a dynamic mechanical analyzer, DMA Q800 (TA Instruments) at ambient conditions. Rectangular samples (30 × 0.5 mm) were heated from 30 to 70 °C at a constant heating rate of 3 °C/min and a frequency of 1.0 Hz.

Electrochemical Characterizations: Cell assembly was carried out in a well-controlled dry room (dew point < -50 °C). High-purity lithium metal foil with thickness of 50 μm (Albemarle, Germany) was used as negative electrode in solid state coin and pouch cells as received. All

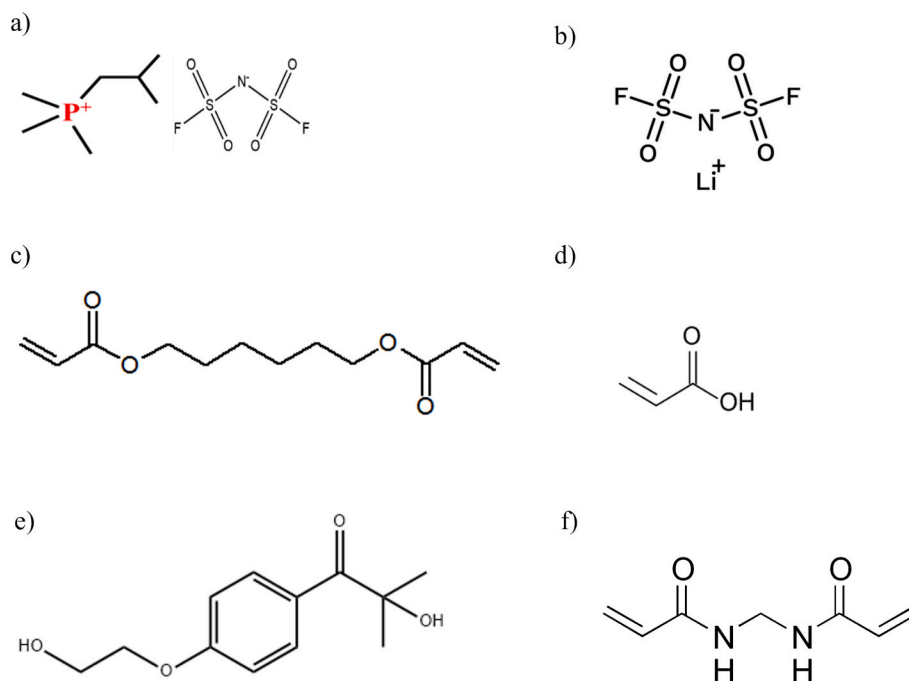


Fig. 1. Chemical structure of (a) P₁₁₁₁₄FSI ionic liquid (b) LiFSI salt (c) 1,6-hexanediol diacrylate (d) Acrylic acid (e) 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (f) N,N'-Methylenebisacrylamide.

electrochemical experiments were monitored using a VMP Biologic potentiostat/galvanostat operated with EC-Lab software except coin/pouch cell galvanostatic cycling, carried out in a BaSyTec CTS Lab multichannel system. Electrochemical measurements were carried out at 50 °C except otherwise stated. These conditions are identical to the cycling conditions used for cells with the non-confined (P111i4FSI ionic liquid) electrolyte (Araño et al., 2020). This elevated temperature was initially chosen to reduce the viscosity of the liquid electrolyte and enhance ionic conductivity. A minimum of two cells were assembled to ensure the reproducibility of the data for all experiments.

2.5. Electrochemical stability window

By using linear sweep voltammetry (LSV) between lithium and stainless-steel electrodes at scanning speeds of 0.1 mV s⁻¹ from OCV to 6.5 V and -1 V vs Li⁺/Li⁰ for the anodic and cathodic branch, respectively, the electrochemical stability window at 50 °C was investigated.

2.6. Ionic conductivity

Electrochemical impedance spectroscopy (EIS) was used to measure the ionic conductivity of the solid polymer electrolyte between two stainless steel blocking electrodes at different temperatures (-20 to 90 °C) in incremental steps of 10 °C. The cells were allowed to stabilize for 2 h at each temperature before the measurement. The measurement range was 184 kHz to 100 mHz with a sinus amplitude of 150 mV. Circular membranes with a diameter of 12 mm and a thickness of 350 µm are used.

2.7. Li/Li symmetric cells

The lithium transference number of the ionogel is determined using a lithium symmetric cell configuration. In this setup, ionogels and lithium-metal electrodes are positioned between two stainless steel spacers which serve as current collectors. The lithium metal electrodes function as both the counter and working electrodes. CR2032 (Hohsen) coin cells were used for this experiment. The lithium transference number is obtained through direct current (DC) polarization, following the method outlined by Bruce and Vincent, where a potential step of 10 mV is applied across the sample, and the resulting current is monitored over time.

For lithium stripping and plating, a similar setup is used. In the variable current density experiment, cycling is performed at different current densities (ranging from 0.1 to 1 mA cm⁻²), with each stripping and plating cycle lasting 1 h. In the constant current density experiment, the symmetric cell undergoes cycling at a fixed current density of 0.3 mA cm⁻² with 1-hour stripping and plating cycles over a duration of 140 h.

2.8. Electrode preparation

A slurry comprising LiFePO₄ (from IBU-tec), poly (vinylidene fluoride) (PVDF, SOLEF 5130 from Solvay) binder, and C65 carbon black (Imerys) in a weight ratio of 95:3:2, respectively, was prepared using NMP as the solvent. The slurry was coated onto aluminum foil (15 µm) by an automatic film applicator 1811/2 (BEVS) and subsequently dried at 120 °C. To enhance electrode density and reduce porosity, the coated electrode film underwent a calendaring process using a 10 cm-diameter roller calendaring machine (DPM solutions). The final electrode has a mass loading and density of ~10 mgLiFePO₄ cm⁻² and ~2 g cm⁻³ respectively. The electrodes were then cut into appropriate sizes (10 mm diameter for coin cells and 59 × 48 mm² rectangles for pouch cells) and subjected to a final vacuum drying step at 120 °C for 16 h to remove residual moisture and optimize battery performance.

2.9. Solid-state battery

Li/ionogel/LFP cells, both CR2032 coin and pouch formats, were manually assembled in a dry room with dew point -50 °C. For Li//LFP coin cells with non-confined (liquid) electrolyte, a 25 µm microporous polypropylene membrane (Celgard 3501, Celgard Inc.) and a borosilicate glass-fiber separator (Whatman GF/D) were used as separators between the electrodes. Single-layer Li/ionogel/LFP pouch cells were manually assembled by step-by-step stacking of lithium metal electrode, ionogel electrolyte and LFP electrode. When cell stack was assembled, nickel and aluminum tabs (MTI Corp.) were ultrasonically welded to lithium and LFP electrode tabs, respectively. Finally, solid-state single-layer pouch cell with an aluminum laminated film soft package (Fig. S1) was sealed under vacuum. Prior to cycling, freshly fabricated single-layer pouch cells were placed between two metal plates, clamped down with torque of 0.3 N m⁻¹ to ensure constant pressure during cycling.

All cells (coin and pouch) were kept for 16 h at 50 °C before cycling to ensure the formation of intimate electrical contacts between both electrodes and ionogel electrolyte. To evaluate cell performance, galvanostatic cycling tests were conducted at a controlled temperature of 50 °C. The cells underwent two formation cycles at 0.05C and 0.1C rates before commencing extended cycling at 0.2C for the coin cells. Single-layer pouch cells were subjected to a single formation cycle at 0.05C followed by extended cycling at 0.1C. All cycling was performed within a voltage window of 2.5 to 4 V.

3. Results and discussion

3.1. Ionogel preparation by in-situ phase separation

Fig. 2 illustrates the fabrication process for the ionogel electrolytes. We employed UV-curing photopolymerization due to its established success in creating ionogels for lithium batteries (Andrzejewska et al., 2009). This method involved UV curing a mixture of an ionic liquid (IL) electrolyte, monomers, crosslinker and a photoinitiator. The monomers used herein are acrylic acid (AA) and 1,6-Hexanediol diacrylate (HDDA) while N,N'-Methylenebisacrylamide and 2-Hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone are the crosslinker and photoinitiator respectively.

It is quite important to know if the polymerization process is complete before further characterizations were performed and also to determine the sufficient irradiation time for UV process. The IR-ATR spectra (Fig. 3) of the liquid precursor showed a characteristic double peak between 1610 and 1640 cm⁻¹, corresponding to the C=C double bond of the acrylic function. This peak however disappeared completely upon 5 min irradiation, confirming complete polymerization and effective UV light diffusion through the precursor.

The optical properties of ionogels provide valuable insights into the solubility of monomers within ILs. As shown in Fig. S2, the AA ionogel appears opaque, indicating the insolubility of AA in the IL. In contrast, the HDDA ionogel is transparent, reflecting the solubility of HDDA in the IL. These optical characteristics also correlate with mechanical properties; soluble monomers produce softer ionogels, while insoluble monomers result in harder materials (Supplementary Video 1). Copolymerization of AA and HDDA yielded a semi-transparent copolymer ionogel, incorporating traits from both individual ionogels. The optical and mechanical properties of the resulting AA-co-HDDA ionogel are influenced by the ratio of the two monomers within the polymer matrix. Preliminary experiments revealed that adjusting the AA-to-HDDA ratio allows the ionogel to adopt properties similar to the dominant monomer. Three compositions were tested: AA:HDDA ratios of 75:25, 50:50 and 25:75. Manual assessment of mechanical properties (as demonstrated in Supplementary Video 1) indicated that the best ability to bend without cracking was achieved with an equal weight ratio of the two monomers. The selected copolymer ionogel consists of IL electrolyte

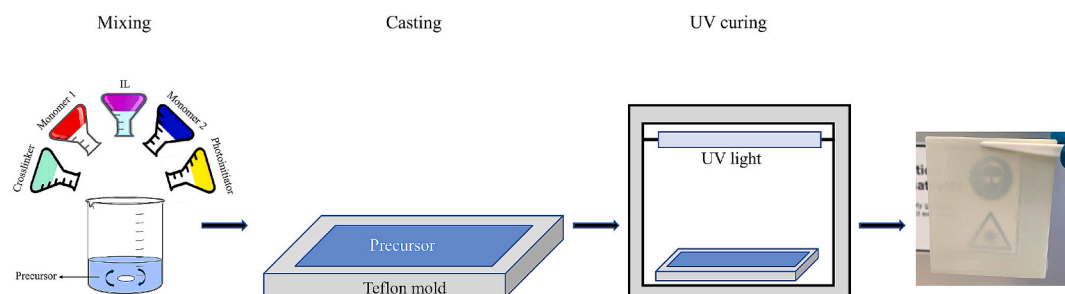


Fig. 2. Schematic representation of the ionogel preparation by UV curing.

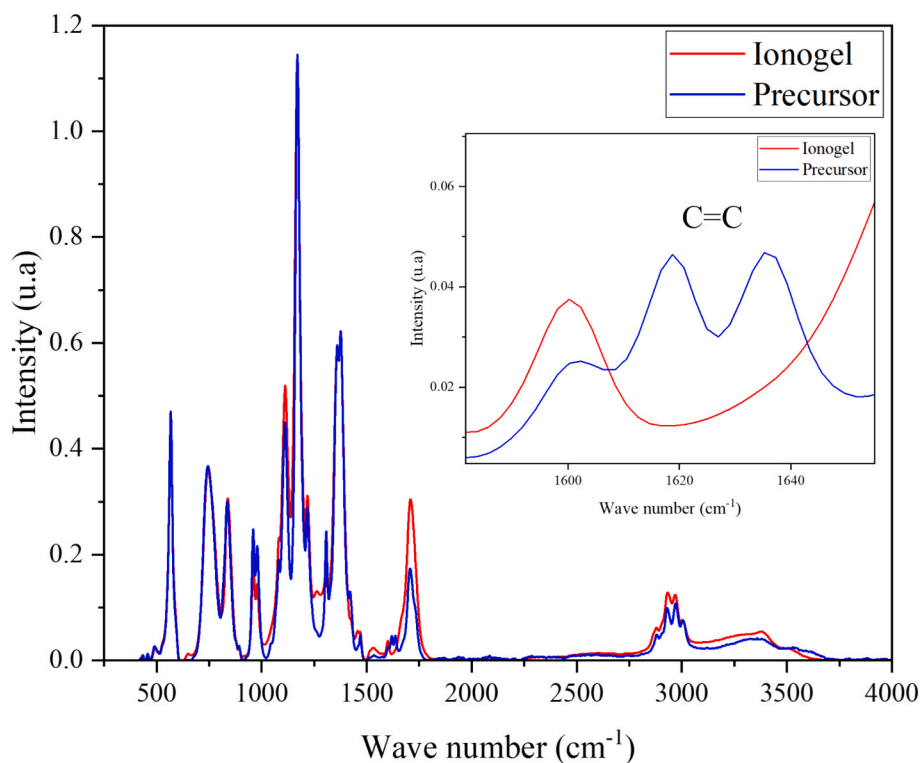


Fig. 3. FTIR spectra of the precursor and resulting ionogel showing the disappearance of the acrylate functionalities upon UV irradiation.

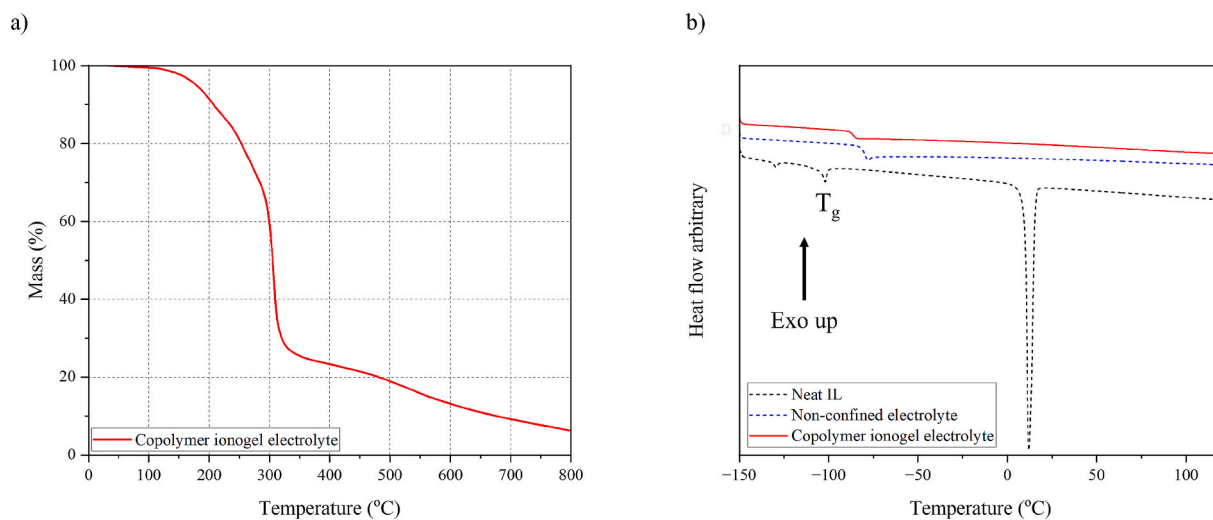


Fig. 4. (a) TGA trace of the ionogel measured under air atmosphere (b) DSC curve of the ionogel showing changes in the T_g upon salt addition and IL confinement under N_2 atmosphere.

(85 wt%), AA (6.5 wt%), HDDA (6.5 wt%), crosslinker (1 wt%), and photoinitiator (1 wt%). This formulation exhibits superior performance compared to individual ionogels, combining the robust mechanical properties of HDDA with the efficient lithium-ion diffusion of AA, as confirmed by ^7Li PFG NMR (Igbaroola et al., 2024).

3.2. Thermal and mechanical behaviour

To assess the thermal stability of the ionogel, thermogravimetric analysis (TGA) was performed under the conditions outlined in the experimental section. Fig. 4a indicated a stable copolymer ionogel structure up to approximately 150 °C, beyond which decomposition of the LiFSI salt commenced, followed by the degradation of AA and HDDA components between 220 and 330 °C (see Fig. S3). Based on these observations, we further investigated the phase transitions associated with the copolymer ionogel electrolytes by Differential Scanning Calorimetry (DSC) experiments.

Since ionic transport is generally faster in poorly viscous liquids, low liquid–solid transition temperatures, which are related to viscosity, are favorable for battery performance. Fig. 4b shows the DSC results for the neat IL, non-confined IL electrolyte (3 M LiFSI in IL) and the copolymer ionogel electrolyte. The neat P₁₁₁₄FSI IL exhibits a melting point at 12.3 °C and a glass transition temperature (T_g) of −101 °C. However, when the IL is doped with 3 M LiFSI, the melting peak disappears, and the system behaves like a supercooled liquid with a T_g of −79 °C. This aligns with previous observations by Girard et al. (2015); who reported the suppression of crystallization and melting peaks at LiFSI concentrations exceeding 2 mol kg^{−1} (~2.6 M) in P₁₁₁₄FSI. They attributed this phenomenon to interactions between lithium ions and the FSI anion. Their work suggests that for electrolyte applications, LiFSI concentrations above 2 mol kg^{−1} in P₁₁₁₄FSI are promising due to the resulting low T_g , enabling wider operational temperature ranges compared to lower LiFSI concentrations. In addition, our study showed that confining the 3 M LiFSI IL electrolyte within the copolymer matrix further reduces the glass transition temperature to −85 °C. This reduction is attributed to the altered molecular dynamics and interactions of the IL in the confined matrix, suggesting an even wider temperature operating window for the ionogel electrolyte compared to the non-confined electrolyte. Furthermore, it provides insight into the enhanced dynamics of the confined 3 M LiFSI IL electrolyte (Demarthe et al., 2024), highlighting the advantages of ionogels for battery applications. Notably, no additional phase transitions were observed between the glass transition temperature and approximately 120 °C, indicating excellent thermal stability within this broad temperature range.

Mechanical properties are critically important for polymer electrolytes especially in the applications of all-solid-state batteries. Fig. S4 shows the dynamic mechanical analysis (DMA) of the ionogel. The ionogel shows a moderately high elastic modulus of about 2 MPa at 30 °C and 1 MPa at 50 °C, which is sufficient for use as a stand-alone polymer electrolyte (Wang et al., 2017). Interestingly, the storage modulus of the membranes is relatively constant between room temperature and 50 °C, probably due to the cross-linking in the material. This is particularly useful for battery operation since it demonstrates that the ionogel maintains its mechanical properties at different temperatures.

Flammability is a critical safety parameter for battery electrolytes, particularly under high current charging/discharging and abusive conditions, where cell temperatures can exceed 80 °C. ILs inherently offer improved safety compared to traditional flammable organic-based electrolytes like LP30. However, the choice of polymer matrix within ionogels can potentially influence overall flammability. To evaluate the safety of the developed copolymer ionogels, flammability tests were conducted. Results demonstrated that the ionogel, like the pure IL, exhibited exceptional resistance to ignition when exposed to an open flame (Supplementary Video 2), unlike the LP30 electrolyte, which readily ignited. These observations underscore the enhanced safety

profile of the ionogel electrolyte, making it a promising candidate for solid-state battery applications requiring high safety standards.

3.3. Transport properties of the ionogel

The ionic conductivity of the copolymer ionogel electrolyte was determined through electrochemical impedance spectroscopy (EIS). By analyzing the Nyquist plot, the ionic resistance (R) was extracted as the intercept of the curve with the real axis. Subsequently, the ionic conductivity (σ) was calculated using the Eq. (1):

$$\sigma = \frac{d}{S \times R} \quad (1)$$

where d represents the sample thickness and S its area.

Fig. 5a presents a comparative analysis of the temperature dependence of ionic conductivity for the non-confined IL and the copolymer ionogel electrolytes. A notable observation is the close proximity of the conductivity values for both systems across the measured temperature range. It is important to remember that the conductivity values of the ionogel represent the conductivity of the entire ionogel, i.e. polymer network (non-conducting) and the confined IL electrolyte (conducting). These values are particularly good compared to some solid-state electrolyte even though the ionogel only contains about 85 wt% IL (Thangadurai et al., 2014). However, a divergence in the slope of both curves becomes apparent at higher temperatures. This slope variation signifies differences in the activation energy (E_a) for ion transport, a critical parameter influencing the temperature dependent ionic conductivity (Demarthe et al., 2024). A steeper slope corresponds to a higher E_a , indicating a higher energy barrier for ion mobility. Conversely, a less steep slope implies a lower E_a , suggesting more facile ion transport. The copolymer ionogel exhibits a less steep slope compared to the non-confined electrolyte, indicating a lower E_a for ion transport within the ionogel. This finding corroborates our previous study on lithium diffusion in AA-based ionogels, which demonstrated a similar trend of lower E_a in the confined state as determined by pulsed field gradient NMR suggesting that interactions at the interface between the IL and the confining polymer matrix enhances Li⁺ transport (Igbaroola et al., 2024).

3.4. Electrochemical characterization of the ionogel

Fig. 5b depicts a typical linear sweep voltammogram (LSV) of the ionogel electrolyte, revealing its very good stability up to 4.2 V vs. Li⁺/Li⁰ (and fair electrochemical stability up to ~5.3 V) at 0.1 mV s^{−1} at 50 °C. This stability aligns with the onset of degradation of the liquid electrolyte (Igbaroola et al., 2024) indicating that the confining monomers do not introduce additional limitations.

3.4.1. Lithium transference number (t_{Li^+}) analysis

The lithium transference number (t_{Li^+}), accounting for the efficiency of Li transport process, is a crucial parameter influencing battery performance. It provides valuable insights into ion transport behavior during battery operation. An increased Li⁺ transference number can reduce the concentration polarization during the charging and discharging processes, thus increasing the energy density of the battery. This experiment was carried out for the copolymer ionogel using a symmetrical Li|ionogel|Li cell configuration by the Bruce-Evans method and the transference number calculated using Eq. (2) (Bruce, 1988).

$$t_{Li^+} = \frac{I_{ss}\{V - I_0R_0\}}{I_0\{V - I_{ss}R_{ss}\}} \quad (2)$$

where R_0 and R_{ss} represent the initial and final interfacial resistances of the cell, respectively; I_0 and I_{ss} are the initial and steady-state currents; and V is the applied polarization potential (10 mV in this study).

The copolymer ionogel exhibited a notable lithium transference

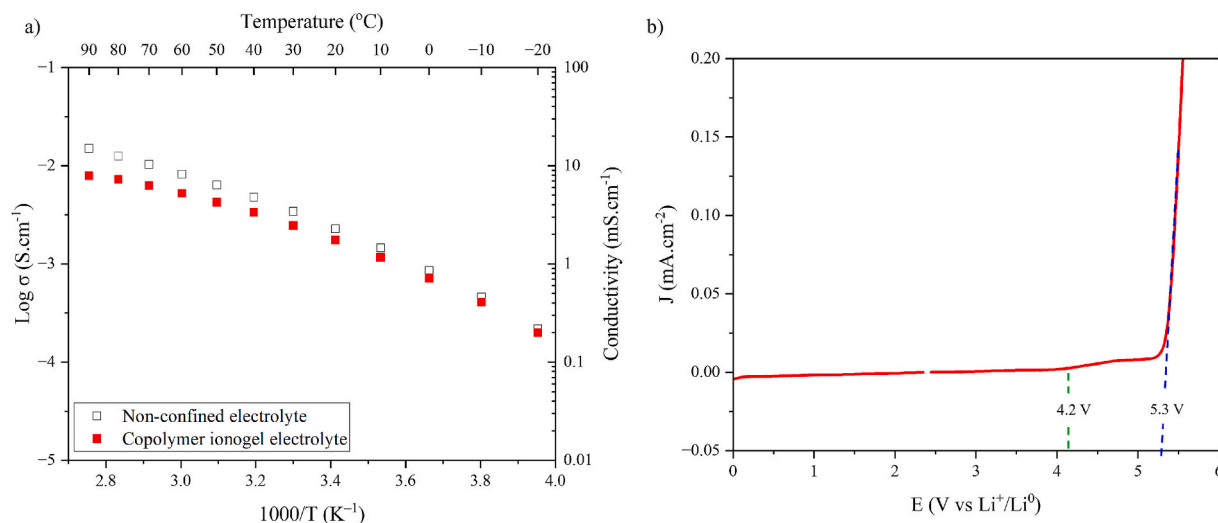


Fig. 5. (a) Temperature dependency of the ionic conductivity and (b) electrochemical stability window of the ionogel electrolyte, respectively.

number (t_{Li^+}) of approximately 0.45, a promising value for a solid polymer electrolyte. In most cases, the Li^+ transference number (t_{Li^+}) is between 0.2 and 0.4 (Li et al., 2024). Fig. S5 shows the current response to the applied polarization potential and the obtained Nyquist plot as a function of impedance measured during the experiment. The Nyquist plots revealed a depressed semicircle with the real axis intercepts corresponding to the bulk electrolyte resistance at higher frequencies and the interfacial resistance associated with surface film layers and charge transfer kinetics at lower frequencies (Peled, 1983; Aurbach and Zaban, 1993). Notably, the observed interfacial resistance was relatively low ($\sim 250 \Omega$), suggesting a good and conductive interface between the ionogel and the lithium metal (Aidoud et al., 2016). This stability is likely attributed to the adhesive properties of the ionogel. Furthermore, the absence of significant variation between the initial (R_0) and final (R_S) interfacial resistances measured before and after DC polarization suggests the formation of a conductive and stable solid electrolyte interphase (SEI) between the ionogel and the Li electrode even though a definitive confirmation should be obtained through impedance measurements at OCV as a function of time (FdZ De Anastro et al., 2019).

3.4.2. Li/Li symmetric cells

The reversible lithium-ion transport properties of the synthesized ionogels were evaluated through symmetrical lithium/ionogel/lithium coin cell cycling. The ionogels were subjected to various current densities ranging from 0.1 to 1 mA cm⁻² for both plating and stripping processes, with five cycles performed at each current density. The resulting polarization profiles, illustrated in Fig. 6a, provide insights into the ionogels' electrochemical behavior. The ionogel demonstrated stable cycling with minimal polarization up to current densities of 0.5 mA cm⁻². At higher current densities, deviations from this behavior were observed, indicative of possible Li dendrites formation. This failure could be attributed to the limitation of the confining polymer matrix as the non-confined (liquid) electrolyte shows good stability up to the maximum tested current density of 1 mA cm⁻² (see Fig. S6). These results establish a safe operational current density limit of 0.5 mA cm⁻² for the copolymer ionogel electrolyte. To assess evolution of polarization upon cycling, the ionogels were subjected to a constant current density of 0.3 mA cm⁻². Fig. 6b shows that the Li/Li cell with the investigated ionogel exhibited stable polarization over 65 charge/discharge cycles, demonstrating its potential for extended cycling applications.

3.4.3. Electrochemical performance of ionogel in solid-state lithium metal batteries

To evaluate the performance of the copolymer ionogel electrolyte,

LiFePO₄ (LFP)-based composite porous electrodes with a relatively high loading of approximately 10 mg cm⁻² were prepared. This loading is closer to that of commercial electrode (~ 15 mg cm⁻²) compared to the lower loadings typically used in previous ionogel studies (Aidoud et al., 2018; Aidoud et al., 2016; Guo et al., 2019; Chen et al., 2018; Liang et al., 2021). The electrochemical performance of a full lithium-metal (Li|ionogel|LFP) cell was investigated. The coin cell (Fig. 7a) exhibited an initial discharge capacity of 160.6 mAh g⁻¹ and a high coulombic efficiency of 98.3 %. This efficiency further improved to 99.9 % in the second cycle and remained stable throughout subsequent cycles. After 60 cycles, the cell retained 97.7 % of its initial capacity. These results are quite comparable to those obtained with the non-confined (liquid) electrolyte. (Fig. S7). It is worth noting that the cells were assembled without IL wetting at the interface between the LFP and the ionogel electrolyte. Oftentimes, wetting is required to enhance electrolyte penetration into the electrode porosity and reduce interfacial resistance. The good contact observed in this study is attributed to the high IL loading (85 wt%). During cell assembly, the applied pressure is expected to induce a slight release of IL at the electrode-ionogel interface, further enhancing the electrode/electrolyte contact and facilitating improved ionic transport. Charge-discharge profiles (Fig. 7b) revealed a slight shift in voltage plateau between the first and second cycles, likely attributed to changes in current rate from 0.05C to 0.1C and then 0.2C. However, the observed polarization remained consistently low (~ 0.1 V taking into account the difference between the charge/discharge profiles), indicating good cell performance and stability over cycling.

The C-rate capability (Fig. 7c) of the Li|Ionogel|LFP cell was evaluated by subjecting it to increasing and decreasing current rates ranging from 0.05C to 2C, with five cycles performed at each rate at 50 °C. The cell exhibited an average discharge capacity of 157.7, 156.6, 152.5, 139.2, 83.0, and 37.0 mAh g⁻¹ at 0.05C, 0.1C, 0.2C, 0.5C, 1C, and 2C, respectively. Notably, the cell demonstrated excellent capacity retention between 0.05C and 0.2C. However, at higher rates (0.5C and above), capacity fade became more pronounced, with 88 %, 50 % and 25 % capacity retention observed at 0.5, 1 and 2C respectively. Upon returning the current rate to 0.05C after 30 cycles, a discharge capacity of only 120 mAh g⁻¹ was obtained, suggesting a degradation of the electrolyte or the electrodes in addition to kinetics limitations. To gain deeper insights into the relationship between C-rate and current density applied to the ionogel, the data was further analyzed as presented in Table S1.

By comparing the rate performance (Fig. 7a) with the current density applied to the ionogel, it becomes evident that operating the cell at current rates exceeding 0.5C pushes the system into a potentially

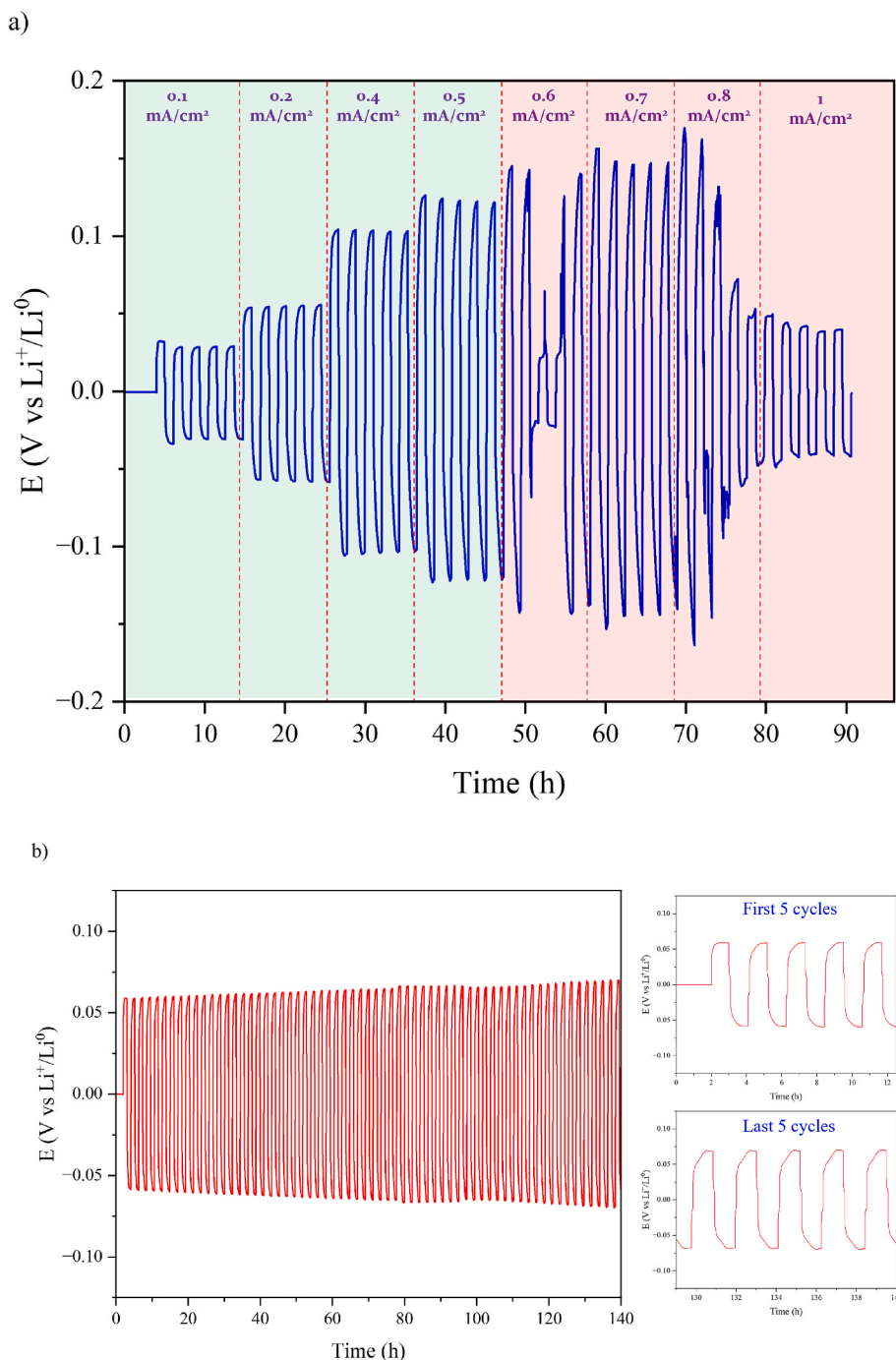


Fig. 6. Polarization profiles of Li/Li symmetric cells with the ionogel electrolytes at (a) variable current densities and (b) fixed current density of 0.3 mA/cm^2 .

unstable region (Fig. 6a). This region is characterized by an increased risk of parasitic reactions or lithium dendrite growth and short circuiting. The charge–discharge profiles (Fig. S8) observed at C-rates higher than 0.5C shows an infinite charging curve suggesting an irreversible reaction, a phenomenon often associated with the formation or growth of lithium dendrites (Homann et al., 2020). Nevertheless, the degradation of the ionogel electrolyte cannot be ruled out. While 120 mAh/g of specific capacity is still observed during discharge, without additional polarization, the drastically lower specific capacity measured on charge for the last cycle along the steep increase of potential seems to confirm the irreversible damage done to the ionogel electrolyte. After this cycle the cell did not function anymore. Therefore, the significant capacity loss observed at 1C and 2C, along with the inability to fully recover

initial capacity upon returning to 0.05C, can be attributed to cell failure due to these adverse electrochemical processes occurring within this high-current density regime.

To assess the scalability of the ionogel electrolyte, lab-size single-layer pouch cell was fabricated. Fig. 8a provides a comparative analysis of the charge and discharge profiles exhibited by CR2032 coin cells and 60×50 mm single-layer pouch cells. While the coin cell exhibited slightly higher initial capacity compared to the pouch cell (resp. 160 mAh/g vs. 154 mAh/g), a corresponding slight increase in polarization was observed for the pouch cell. This is likely attributed to the increased complexities involved in pouch cell fabrication, including gas evolution during cycling, which could affect the internal pressure within these cells. However, the extended cycling performance (Fig. 8b) of both

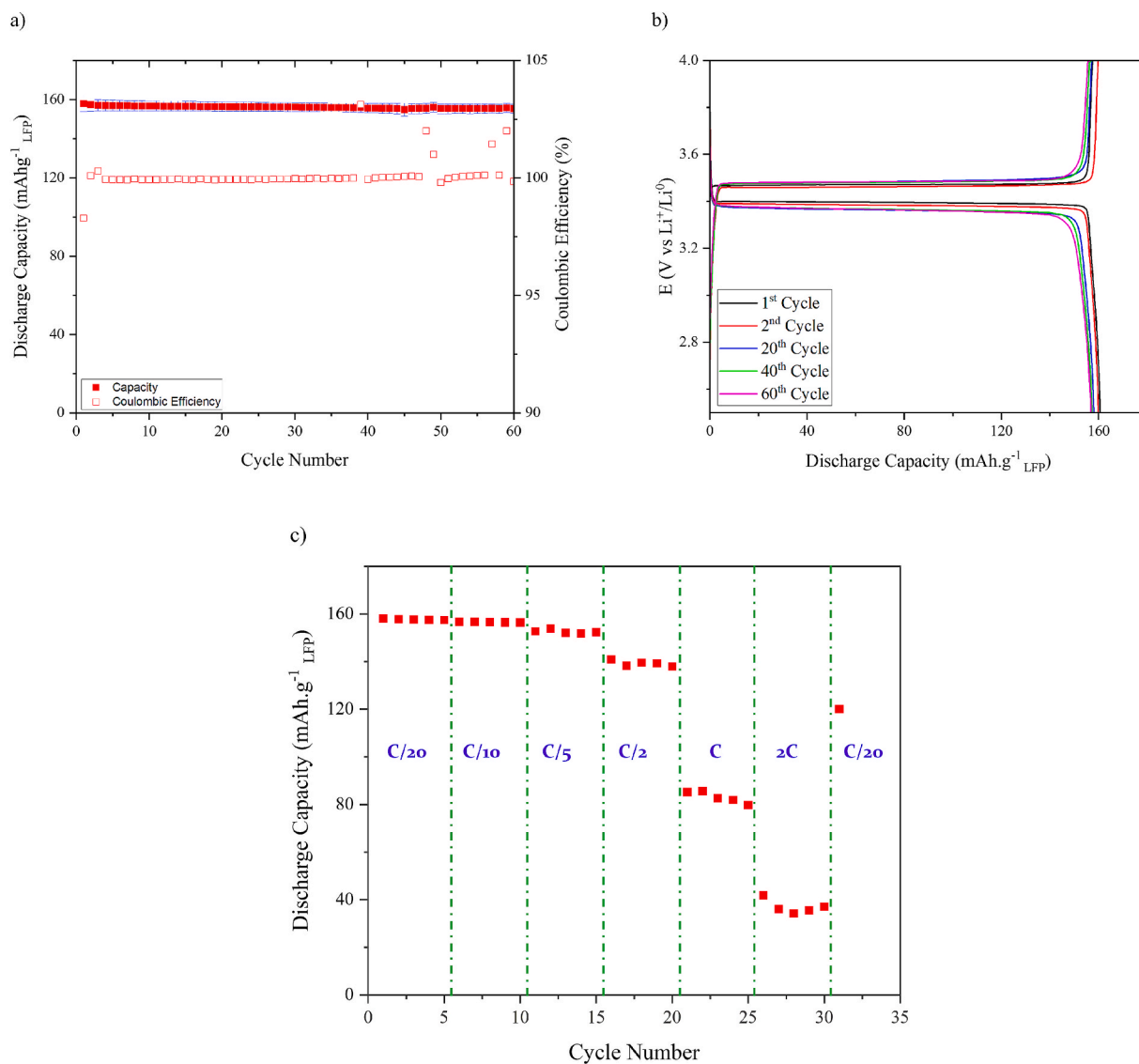


Fig. 7. (a) Cyclability of the ionogel as electrolyte in Li|ionogel|LiFePO₄ battery (b) charge and discharge profiles of the same battery at selected cycles (c) C-rate capability performance of the ionogel electrolyte in Li|ionogel|LiFePO₄ battery.

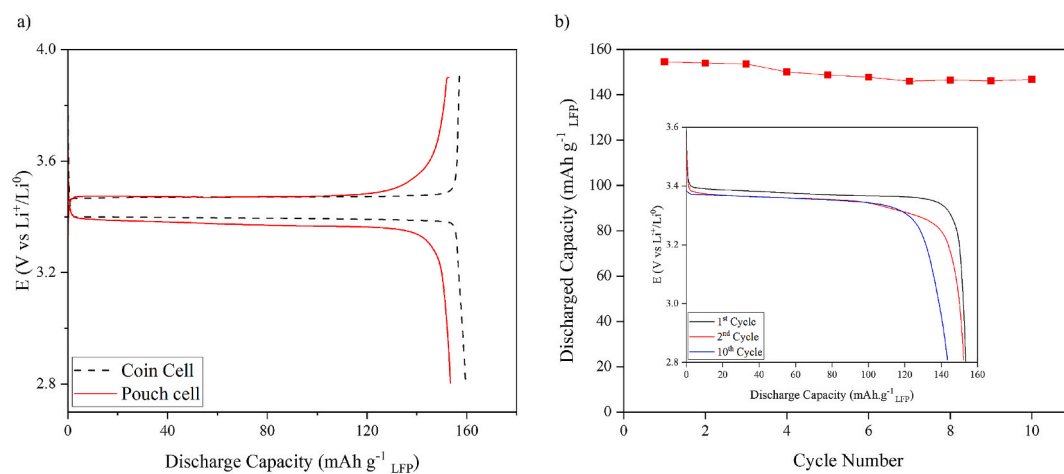


Fig. 8. (a) First cycle charge and discharge (0.05C) profiles comparison between coin and pouch cell formats of Li|ionogel|LiFePO₄ cells cycled at 50 °C (b) Cyclability of the Li|ionogel|LiFePO₄ pouch cell and discharge profiles of the same battery at selected cycles.

cell formats demonstrated comparable electrochemical behavior. To further optimize pouch cell performance and potentially achieve capacity values closer to those of coin cells, refinements in the pouch cell fabrication process are necessary. This should include impregnating the electrode with the ionogel precursor in the form of a catholyte to improve cell resistance.

4. Conclusions

In this study, we developed a tough, flexible, and self-standing copolymer ionogel electrolyte via a straightforward in situ phase separation method. This approach combines the favorable lithium-ion transport characteristics of acrylic acid with the mechanical robustness of 1,6-hexanediol diacrylate. By incorporating 85 wt% of an IL electrolyte (3 M LiFSI in P₁₁₁₄FSI) within an AA-co-HDDA copolymer matrix, we achieved a high ionic conductivity ($\sim 2.5 \text{ mS cm}^{-1}$ at 30 °C), comparable to the non-confined IL, along with excellent lithium transference number and very good stability up to 4.2 V vs. Li⁺/Li⁰ (and fair electrochemical stability up to ~ 5.3 V). Additionally, the ionogel exhibited enhanced thermal stability and inflammability compared to traditional liquid electrolytes based on organic solvents. Lithium-metal battery (Li|ionogel|LiFePO₄) testing showed a capacity retention of $\sim 98\%$ of the initial capacity after 60 cycles with a stable polarization. The successful fabrication and testing of a lab-scale pouch cell further demonstrated the scalability of this ionogel electrolyte. Given its combination of high performance, safety, and ease of fabrication, this ionogel holds significant promise for commercial solid-state lithium metal battery applications. Future research will focus on long-term cycling investigations to assess the durability and performance degradation of the cells. Additionally, post-mortem analyses will be conducted to elucidate the underlying failure mechanisms and structural changes occurring during cycling.

CRedit authorship contribution statement

Boluwatife Igaroolo: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nicolas Demarthe:** Validation, Methodology. **Patrick Howlett:** Writing – review & editing, Methodology. **Maria Forsyth:** Writing – review & editing, Validation. **Galyna Kvasha:** Visualization, Validation. **Andriy Kvasha:** Validation, Methodology. **Susan Sananes-Israel:** Validation. **Iratxe De Meatza:** Writing – review & editing, Validation, Supervision, Resources. **Nicolas Dupré:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jean Le Bideau:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: DUPRE Nicolas reports financial support was provided by Horizon 2020 European Innovation Council Fast Track to Innovation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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pouch cells.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2025.121965>.

Data availability

Data will be made available on request.

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